

MTH00050 – Combinatorial Mathematics

Lecture 0: Introduction to Combinatorics

Lecturer: Bùi Văn Thạch

Lab instructors: Nguyễn Ngọc Toàn, Trần Thị Thảo Nhi, Lê Đức Khoan

{bvthach,nntoan,tttnhi,ldkhoan}@fit.hcmus.edu.vn

Outlines

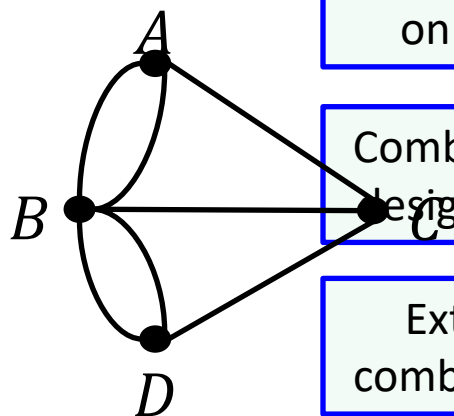
- Course information
- Course topics
- Grading
- Course requirements
- References

Outlines

- Course information
- Course topics
- Grading
- Course requirements
- References

15 branches of combinatorics

the study of graphs, which are mathematical structures used to model pairwise relations between objects



Algebraic
combinatorics

Graph theory

Enumerative
combinatorics

Topological
combinatorics

Probabilistic

the number of ways that certain patterns can be formed.

COMBINATORICS

Combinatorics
on words

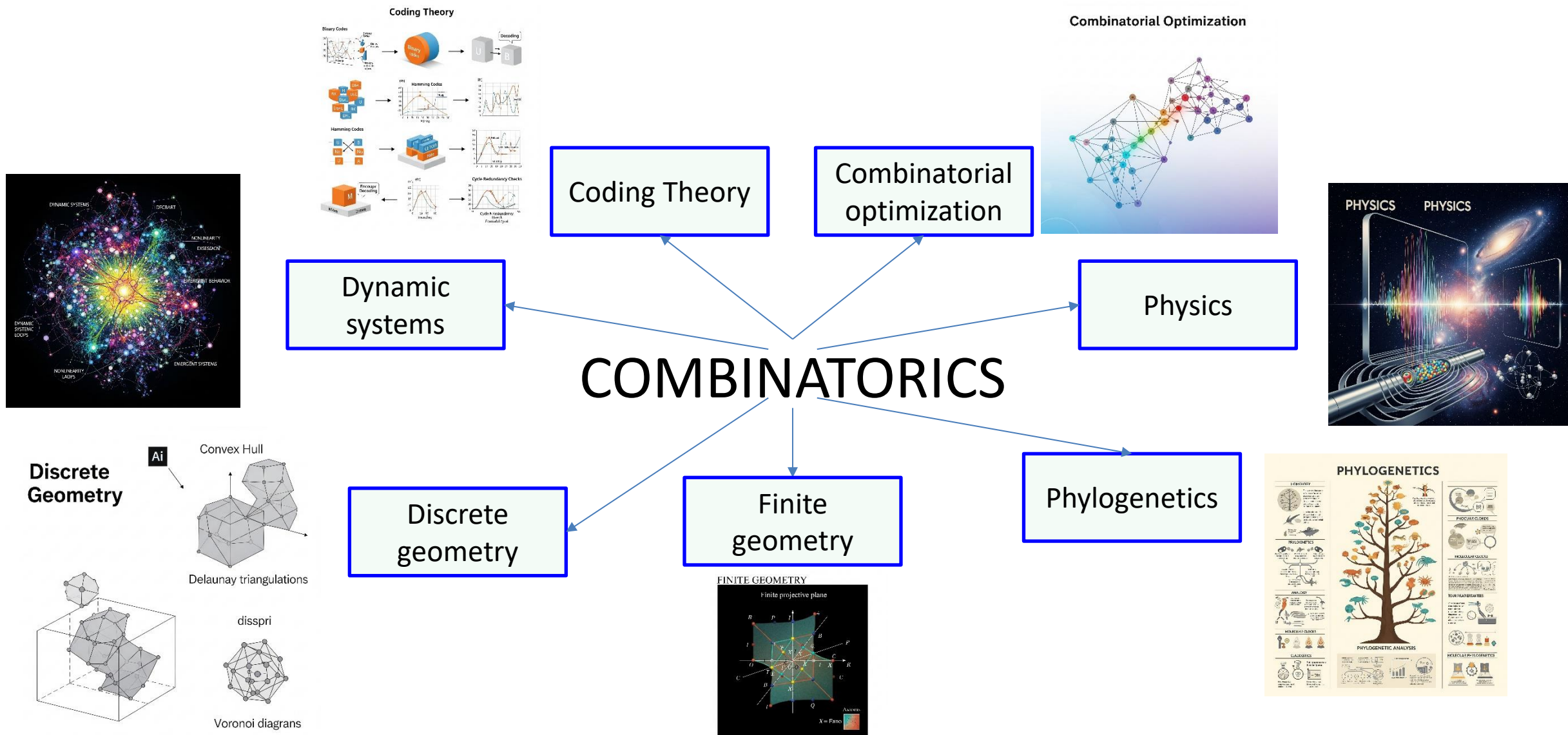
Combinatorial
design theory

Extremal
combinatorics

Geometric
combinatorics



Multi-disciplinary fields that include combinatorics



Enumerative Combinatorics: Learning Outcomes

1. Understand basic counting principles:
Apply the Addition and Multiplication Principles.
2. Use advanced counting techniques:
Solve problems using permutations, combinations, and binomial coefficients.
3. Can solve basic recurrence relations.
4. Use generating functions:
Construct and manipulate ordinary and exponential generating functions for enumeration.
5. Use inclusion-exclusion principle and Mobius inversion:
Count the predefined problems.
6. Understand combinatorial identities and numbers, such as Catalan and Stirling, to exactly or approximately derive the values of equations.

Graph Theory: Learning Outcomes

1. Be fluent in the language of graphs so you can analyze and model problems effectively.
 - Understand foundational terms: vertex, edge, degree, path, cycle, connectedness, adjacency.
 - Distinguish between types of graphs: directed, undirected, weighted, unweighted, cyclic, acyclic.
2. Choose the right data structure for a given graph problem to optimize performance.
 - Adjacency matrix, Adjacency list, Edge list.
3. Explore, analyze, and manipulate graph structures.
 - Depth-First Search (DFS),
 - Breadth-First Search (BFS).
4. Learn to find Eulerian paths, Eulerian circuits, and the hardness of finding Hamiltonian circuits.
5. Learn how to find spanning trees and shortest paths.
6. Learn some problems in network flows.

Outlines

- Course information
- **Course topics**
- Grading
- Course requirements
- References

Course topics

0. Introduction
1. Sets
2. General counting methods for arrangements and selections
3. Inclusion-exclusion principle
4. Recurrence relations
5. Generating functions
6. Combinatorial identities and numbers:
Catalan and Stirling
6. Graphs
7. Path problems
8. Tree problems
9. Network flows

(Tentative) Schedule 2026

No.	Date	Location	Topic
1	13/01 16/01	C24 C24/E302	Introduction, Sets, Proofs
2	20/01 23/01	C24 C24/E302	General counting methods
3	27/01 30/01	C24 C24/E302	Inclusion-exclusion principle
4	03/02 06/02	C24 C24/E302	Recurrence relations
5	03/03 06/03	C24 C24/E302	Generating functions
			Mid-term test

No.	Date	Location	Topic
6	10/03 13/03	C24 C24/E302	Graphs (Part I)
7	17/03 20/03	C24 C24/E302	Graphs (Part II)
8	24/03 27/03	C24 C24/E302	Path problems
9	31/03 03/04	C24 C24/E302	Tree problems
10	07/04 10/04	C24 C24/E302	Network flows (Part I)
11	14/04 17/04	C24 C24/E302	Network flows (Part II)
			Final exam

Outlines

- Course information
- Course topics
- Grading
- Course requirements
- References

- Participation: 5%
- Class activities: 5%
- Lab: 20%
- Mid-term test: 30% (Closed exam)
- Final test: 40% (Closed exam)
- Cheating: getting 0 for the final result.

Outlines

- Course information
- Course topics
- Grading
- Course requirements
- References

General rules (1/2)

- Do not disturb others in class.
- To be on time and actively participate in class activities.
- There are some quizzes during the course.
- Keep your phone in silent mode.

General rules (2/2)

- Students must strictly adhere to the regulations and rules of the Faculty and the University.
- Students will be barred from taking the final exam if they were absent more than **3 times** in total for theoretical classes.
- **Zero for the final exam or the lab** will result in **failure of this subject**.
- In cases of dishonesty during assignments or exams, students will face all disciplinary actions from the Faculty/University and will receive a score of **0** for THE COURSE.

Chatbot policy (1/2)

- With the advent of AI-driven chatbots like ChatGPT, Gemini, and others, there's a new dimension to the academic resources available to students. These tools can be instrumental in fostering understanding, but they come with their own set of guidelines for responsible use:
 - Using chatbots for enhancing clarity on a topic or concept is acceptable.
 - Directly copying and pasting responses or solutions from these chatbots without a genuine understanding is deemed a severe breach of academic integrity.
 - Should you take assistance from chatbots, it is imperative to explicitly acknowledge it in your submission.
- Non-compliance with the above directives, akin to other forms of academic misconduct, will be treated with utmost seriousness, resulting in consequences as outlined above.

Chatbot policy (2/2)

- If AI usage $\geq 70\%$, the maximum score is 60/100.
- From 50% to $< 70\%$, the maximum score is 80/100.
 - If the student who claims this cannot answer 3/5 questions, misconduct (lying) is detected.
- If $< 50\%$, the maximum score is 100/100.
 - If the student who claims this cannot answer 2/3 questions, misconduct (lying) is detected.

Lab Sessions

- Follow the guidance of the teachers.
- Do not hesitate to ask questions.
- Try your best to get as much experience as you can.
- Language: C++/Python.
- IDE: Visual Studio.

Other notes

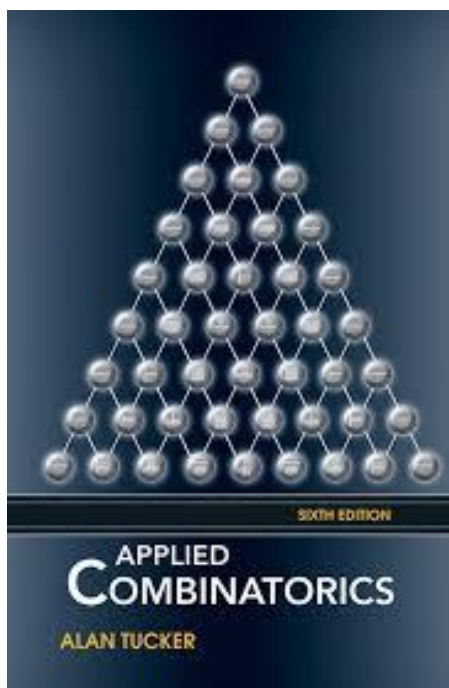
- Use official email always.
- Read textbooks more than the requirements.
- Get the knowledge from the videos suggested by the instructors.

Outline

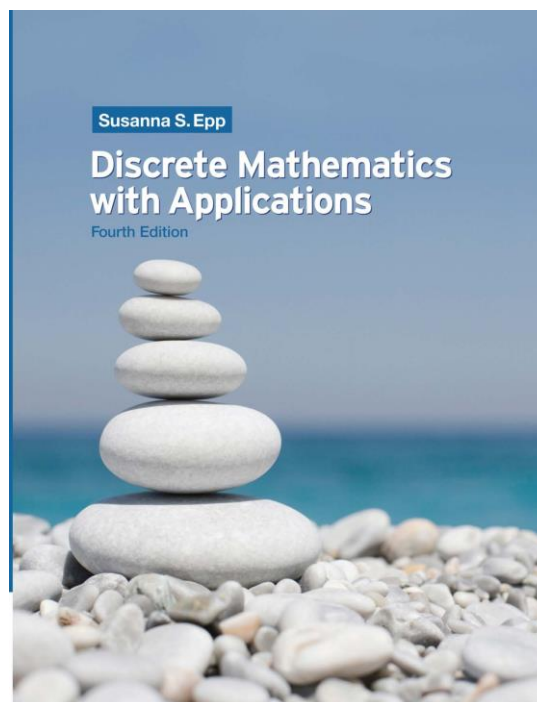
- Course objectives
- Course content
- Student duties
- Grade evaluation
- References

Main references

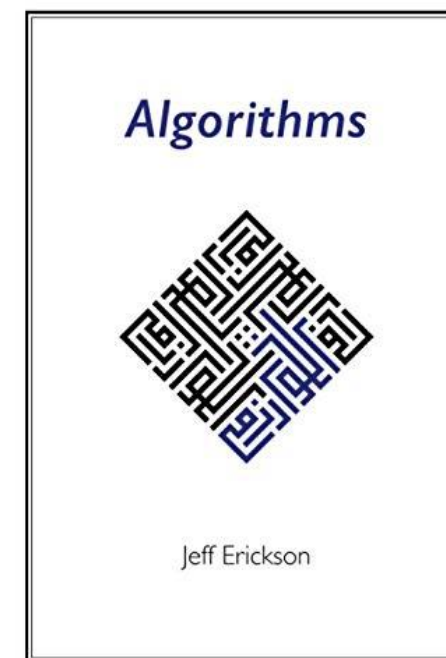
[1] Alan Tucker, Applied Combinatorics. 6th edition, JohnWiley & Sons, Inc, 2012.



[2] Susanna S. Epp. *Discrete Mathematics with applications*, 4th edition, Brooks/Cole, Boston. 2011.

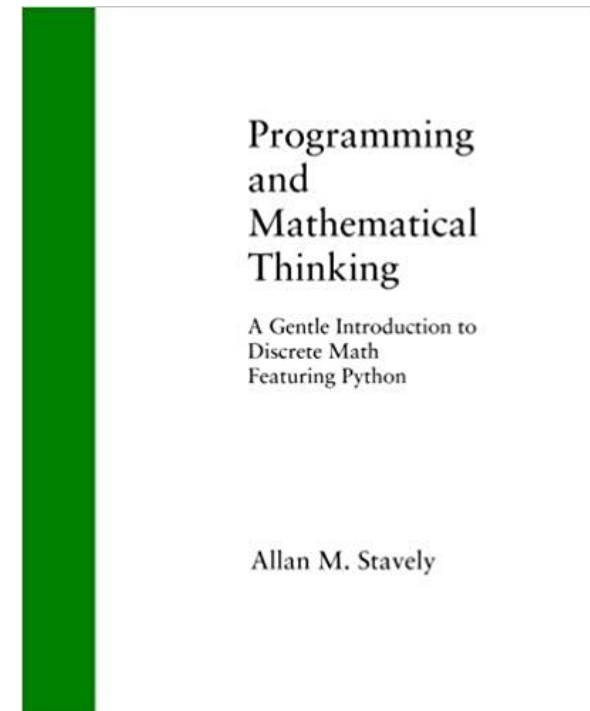
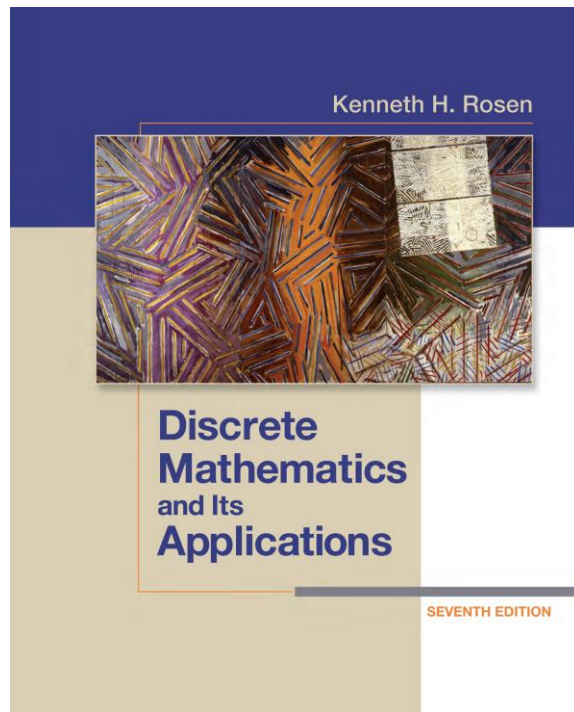


• [3] Erickson, Jeff. *Algorithms*. Independently Published. 2019.



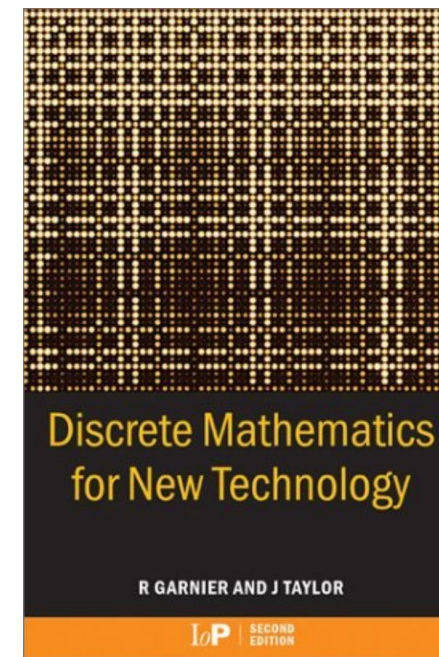
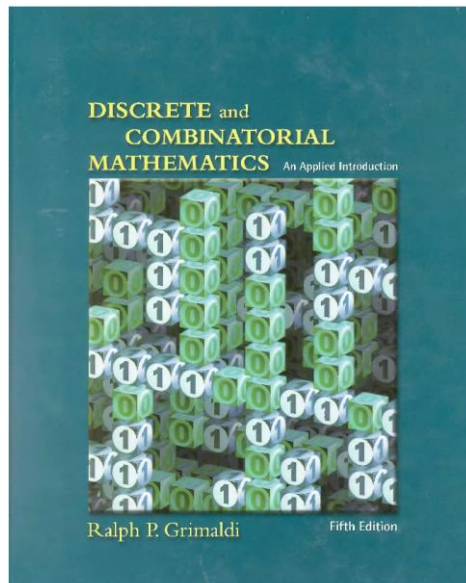
Recommended references

- [4] Kenneth H. Rosen. *Discrete mathematics and its applications*, 7th edition, McGraw-Hill, New York. 2012.
- [5] Allan Stavelly. Programming and mathematical thinking: A gentle introduction to discrete math featuring Python, The New Mexico Tech Press, Socorro, New Mexico. 2014.



Other readings

- [6] R. P. Grimaldi. *Discrete and Combinatorial Mathematics: An Applied Introduction*. 5th edition, Pearson Education, Boston. 2004
- [7] Rowan Garnier, John Taylor. *Discrete Mathematics for New Technology*. 2nd edition, Institute of Physics, Bristol, Philadelphia. 2002.



Other readings

- [8] Trần Ngọc Danh, Toán rời rạc nâng cao, NXB ĐHQG-TPHCM, 2001.
- [9] Dương Anh Đức, Trần Đan Thư, Lý thuyết đồ thị, NXB ĐHQG-TPHCM, 2008.
- [10] Walter D. Wallis, John C. George, Introduction to Combinatorics, 2nd Edition, Chapman and Hall/CRC, 2017.

Q & A